

Working Session on Technical Assistance for the Housing Reconstruction NRA Conference Hall, 2nd December 2016

Participants: ASF Nepal, BBC Media Action, Build Change, CARITAS (Germany), Concern Worldwide, CRS, DFID, Embassy of India, Finn Church Aid (FCA) Nepal, Habitat for Humanity, HELVETAS, ICIMOD, IFRC, British Red Cross, Japanese Red Cross (JRCs), JICA, Medair, NSET-Nepal, OXFAM, Tearfund Nepal, UNDP, and UNOPS.

Introduction

Dr. Chandra Shrestha and Ram Chandra Shrestha of the Housing, Local Infrastructure, and Settlement Development division of the National Reconstruction Authority (NRA) opened the meeting. They informed that previously the division had been focused on rural housing, but this has now been changed and they are now responsible for rural and urban housing. Earlier in the year the division had been planning to have coordination meetings with Partner Organisations (POs) every month, but eventually these did not go ahead as other sections of the NRA were organising similar meetings and they did not want to overlap. They presented the objectives of the working session which were for the NRA to express expectations and needs for implementation of technical assistance, to have an opportunity to learn from best practices of POs already implementing, and to develop a modus operandi for moving ahead with future objectives.

Baliyo Ghar – A Programme to Ensure Build Back Safer in Earthquake Reconstruction, Rajan Dhungel, NSET

NSET have been implementing technical assistance through the 'Baliyo Ghar' programme for 15 months, and to date:

- 1,074 orientations on the housing reconstruction, with 30,152 participants, have been held,
- 1,468 masons have been trained,
- 556 instructors for masons training have been trained, and
- More than 1,500 house construction sites have been reached with supervision.

Baliyo Ghar technical assistance is long term with the project running until September 2020. The programme is providing TA across one third of VDCs in Dhading, Nuwakot, and Dolakha and also in one municipality in Kathmandu. NSET are also providing support to POs on training of staff, technical advice, and establishing a National Reconstruction Technology Centre (NRTC) to spread the influence of NSET beyond the Baliyo Ghar programme.

The ultimate goal of Baliyo Ghar is support households to build safer and disaster resilient houses. The programme has set a target that a minimum of 70% of houses constructed (not just those damaged by earthquakes but all built in the area) will be compliant. The aim is reach 100% but the minimum achieved will be 70%. In order to ensure this, Baliyo Ghar is delivering masons training, conducting orientations to households, and providing mobile technical assistance teams across the programme's working areas. A key principle of the Baliyo Ghar programme is a blanket approach to providing TA, with construction support and advice provided to households whether or not the house was damaged by the earthquake.

The Baliyo Ghar implementation approaches at national, district, and local level were presented. At national level Baliyo Ghar provides technical support on the development of curricula, guidelines, and manuals, and also conducts advocacy on key technical issues. This is primarily done through the National Reconstruction Technology Centre (NRTC). Baliyo Ghar is also developing instructors for other districts. At district level the District Reconstruction Technology Centre (DRTC) is the base for the project implementation with the project support teams based there and training programmes running in the centre. At village level, Local Reconstruction Technology Centres (LRTCs) are established to provide backstopping for mobile technical teams, to be a training centre, and to be centre point for the information and awareness campaigns. The setup of the team at each level was also presented in terms of number of staff and staff profiles.

Baliyo Ghar has also supported the revision of the GoN training curriculum for existing masons so that it is now split into a 7 day course for rural masons and a 7 day course for urban masons rather than one combined course. Research is also being conducted to look at different construction options for rural areas, as previously presented at the HRRP technical session on [‘Enhancing Rural Construction Culture’](#).

On the Job Training is very important as there are very few masons that can be trained under the 7 day curriculum. New masons are needed to meet the demand of the households during reconstruction. Awareness activities, especially mobile technical assistance teams providing door to door assistance, need to be implemented alongside training for the training to be effective. Different instructor development courses are also being delivered by Baliyo Ghar. To become a full instructor, the engineer must complete the ToT, then observe a training, then contribute as an assistant trainer, then deliver a training under supervision, then complete the Training for Instructors (Tfi) course. Scattered trainings can create confusion at community level and there is a need for a standard approach. People need technical advice during construction at the right moments and this will come through the mobile technical assistance teams.

[Technical Assistance by JICA, Hiroshi Imai, JICA](#)

The JICA approach to technical assistance is quite similar to NSET’s Baliyo Ghar programme with activities at field level focused on training and mobile technical assistance teams, and at central level providing support for the development of the DUDBC design catalogue, various policies, and other technical guidance.

JICA’s field level activities are implemented in Gorkha and Sindhupalchowk across a total of 38 VDCs. To date, JICA have trained 1,431 masons, 3,647 house-owners, and trained 440 engineers (including sub-engineers and assistant sub-engineers) as trainers. JICA also has mobile teams in place surveying new construction quality, providing technical information, and conducting awareness raising activities.

At central level, JICA’s support for the development of the DUDBC design catalogue included designing 6 models for and completing structural analysis for all the models. JICA have also developed the minimum requirements for load bearing structures; stone and mud mortar, stone and cement mortar, brick and mud mortar, and brick and cement mortar. JICA are now developing a handbook on the minimum requirements which has been submitted to the NRA for review and approval. This will be used to disseminate minimum requirements in the field, and two posters have also been developed with one on the construction sequence and one on the minimum requirements.

JICA have also supported the development of an inspection manual to accompany the [inspection SOP](#). This manual provides step by step guidance on how to implement the SOP and teaches the DL-PIU how to fill in the forms and follow the SOP procedures. A two day training programme for DL-PIU engineers on inspection has been developed and will be delivered in the districts very soon. Refresher training for masons is also planned.

JICA is now supporting DUDBC to develop the second volume of the design catalogue to promote options for new construction techniques and methods. There are also issues with emerging technical approaches such as houses where the ground floor has been constructed in masonry and the upper floor is constructed in timber and CGI. This has been found in lots of places but there are no minimum requirements for this type of construction. As it would be so time consuming to develop the minimum requirements for this type of house it is suggested that it would be more effective to start with developing fixed design for new techniques such as this.

JICA are planning to support the development of a ‘correction manual’ to support how non-compliant construction will be addressed. This includes research on retrofitting approaches that could be implemented.

[Noll Tufani, Build Change](#)

Build Change are strong supporters of home-owner driven construction as it reduces the logistics required, increases efficiency, leads to job creation, is a durable and sustainable investment, and ensures that the building practices developed during the reconstruction are replicable in the community. Specifically in Nepal, Build Change is implementing technical assistance as follows:

- **Builder training:** more than 500 builders have been trained to date. This is seen as a key prerequisite for TA; there is no stock of skilled builders any other TA will not have any impact. Build Change conduct two types of training, one is a refresher for skilled builders and the other is On the Job Training which develops new builders.
- **Community awareness:** it's very difficult to provide targeted technical assistance in a community that isn't ready to receive the information. One type of community awareness that Build Change use is forum theatre. This is a play based on the Shelter Cluster 10 key messages and it lasts 45 minutes. It is an interactive event within the community. To date the forum theatre has been delivered in 6 districts and nearly 30,000 people have seen it. Another community awareness activity is the construction of model houses. This is also linked to On the Job Training. The model houses allow people to see something concrete, and they become a beacon of safe construction. The model houses have colour coded earthquake resistant construction element and these elements are explained on a bill board placed at the house. Once the community awareness raising activities have been carried out and skilled labour is in place, then Build Change move to conducting targeted TA for home-owners.
- **Engineering design support:** technical support centres have been established in 7 VDCs across 5 districts. The centres provide 'Build Back Safer' information and the centre teams conduct site visits from there. Free of charge design services are also provided so that households can get DUDBC approval for their designs. A mobile app called '[Surakshit Ghar](#)' has been developed as an extension of those centres. The app can be downloaded for free from the google play store and is all in Nepali. It provides information to communities on key GoN information that has been published, e.g. info on enrolment, accessing the housing grant, the DUDBC catalogue of designs, and also provides names and contact details of trained builders by VDC. The contact details of the nearest technical support centre are also provided and the app includes a feature which allows households to take a picture of a situation and send this to the technical support centre. The centre staff can then respond by phone, conduct a visit, or invite people to come in. There is a mobile team associated with each centre but the app is designed to allow for broader reach and more targeted interventions.
- **Construction supervision and monitoring:** Build Change have developed another mobile app for site supervision and monitoring. To date this app includes 2,100+ houses and 1,100 site visits have been recorded. The app allows for site surveys to be linked back to the households Project Agreement (PA) and also allows for collection of socio-economic data about the households. Whilst Build Change cannot do the inspections, they have been using the site survey data collected through the app to provide recommendations to DUDBC regarding whether or not the second tranche should be released.

Technical Assistance Working Session, Loren Lockwood, HRRP

From the previous presentations it is apparent that there is quite a range of activities that can be considered Technical Assistance (TA). The HRRP has been analysing the 4W data and partners' responses to a survey on this to better understand what activities POs are doing and what the quality is of those activities. Based on this the HRRP broadly came up with a set of six activities to be considered as the full package for technical assistance:

- Community / Household level orientations
- Mobile teams / supervision
- Technical support helpdesks / resource centres
- Demonstration construction
- Existing masons training – 7 day DUDBC Curriculum
- New masons training – 50 day CTEVT Curriculum

A map of the level of coverage of these activities was presented (available [here](#)). The darker the area, the higher the number of activities that is being implemented there. There are only 9 VDCs that have full coverage of these activities and there are dramatic gaps. The negative picture of the first map was also presented (available [here](#)). This shows the areas where there is no technical assistance in relation to damage levels, where the darker shades of red indicate higher damage levels. There are enormous gaps in coverage of technical assistance in areas where damage levels are

very high and people are already starting to rebuild. It is also clear that in areas where damage levels are lower, due to the gaps in coverage they have no access to information on what options may be available to them, especially retrofitting.

The gaps in the provision of TA are not just due to lack of coverage, but also related to the quality and amount of TA that is being provided. It is essential to remember that TA is an ongoing process and that one visit to a community is not sufficient. TA must be in place until the last house has been rebuilt and must be present every step of the way. Integration with, and support to, the DL-PIU engineers at VDC level will be extremely important. There are also a number of agencies that are only partially covering a VDC, leaving hard to identify gaps and creating social issues through inequitable support. There are also many places in areas of high damage where agencies are keen to work and duplication is becoming an issue. Without proper coordination with the NRA, DUDBC, and HRRP at national and district level, duplications are hard to pick up. There are now areas where the 7 day training for existing masons has been delivered to 850% of the masons in a VDC. There are also areas where no masons training is taking place. It is up to all of us to get together to work to rectify this situation and ensure a more even level of coverage.

Another challenge is the quality of TA that is provided. There is a distinct lack of standardisation of materials, with a large set of materials rolled out based on the [Shelter Cluster 10 key messages](#) which do not cover all the elements for compliant construction under the [inspection SOP](#). There are also different definitions of activities with some organisations reporting that they do household orientations, where the orientation is one visit for half an hour to the whole community, and others reporting that they do household orientations, where the orientation is a 4 hour session at household level and it is repeated 4 time over the course of 3-6 months. These are very different activities, and the HRRP is now working to revise the [4W](#) to try to avoid this reporting issue, but agencies may also need to look at the way they are defining activities. In terms of the quality of training there are also issues with trainers not going through a trainer development process. Completing the ToT is only the first step and beyond this new trainers need to have opportunities to observe trainings, participate as an assistant trainer in trainings, and act as lead trainer under supervision before they can independently deliver trainings. The issues with quality and coverage of TA are leading to large discrepancies in the investment in VDCs with some receiving large amounts of support across the whole VDC, in some VDCs only some wards are receiving support, and in some VDCs there is no support.

To try to resolve the coverage and quality issues it is [proposed to establish minimum requirements for technical assistance](#), and one of the aims of the session was to contemplate this as a group. The first step in defining the minimum requirements is to look at what the objective of TA is. The major objectives were presented as access to timely and appropriate advice, availability of skilled construction workers, and developing community resilience. There are a range of activities that can be used to achieve these objectives. The HRRP have developed a [discussion paper](#), which has been shared with you all, as a first step in proposing the minimum requirements for TA.

Demonstration construction may now take a different form because of the On the Job Training approach being recommended by the NRA. This approach involves construction of one demonstration house per ward which should be used for training new masons through the On the Job Training, 50 day CTEVT curriculum. The demonstration houses can be built for a household, where the household must be selected by the village reconstruction committee and the household must be able to cover all costs above the 300,000 NPRs housing grant but the labour cost can be covered by the implementing agency. The demonstration houses should also be used to provide a practical example of better construction practices for the whole community over the course of the reconstruction. A map of the coverage of vocational training (based on the NRA targets) was presented (available [here](#)) and it is clear that there is an enormous gap across the 14 districts, and this continues into the 17 districts. A map of coverage of the training for existing masons was also presented (available [here](#)) where coverage is presented based on an estimate that there is approximately 1 existing mason for every 20 houses. This estimate was based on a survey conducted in Gorkha and then extrapolated for all 14 districts in order to estimate coverage. Based on this there is a very high level of duplication. Most of the colours on the map indicate more than 100% coverage and the darker colours indicate 500-900% coverage. Given the

high level of gaps, duplication should not be happening. There is a need to have better reporting and coordination at district and national level, and gaps as priorities rather than areas where we are working specifically.

There is sufficient financing in place for quality TA to be delivered at scale across the 14 districts, and also the 17 districts. The GoN has the finances to provide the housing grant, and POs are really urged to focus on TA and work to cover the huge gaps.

As a next step the HRRP is adjusting the 4W to reflect the minimum requirements and it is hoped that this will improve reporting and ensure a better understanding of quality and coverage of TA. The HRRP also strongly urged all partners to sincerely consider stretching their level of coverage to try to better address the huge gaps, the partial coverage of VDCs, and to avoid duplication. Partners were also urged to conduct internal checks on the quality of activities delivered. The need for TA to start to roll out in the Kathmandu Valley was also raised.

Questions and Discussion

NRA: The discussion around the overwhelming gaps and overlaps was initiated by the NRA in April 2016, and efforts were made to resolve this but unfortunately it wasn't possible at that time. POs were already associated with places and it didn't seem possible to expand or change working locations. The NRA repeatedly requested that the HRRP support with this but the previous incarnation of the HRRP repeatedly showed coverage is fine. Now that there is clarity on the fact that there are real genuine gaps and overlaps the NRA urged POs to support this issue to be resolved or at least minimised. The NRA requested that the focus be on two key areas:

- **On the Job Training:** In order to resolve gaps in On the Job Training (OJT) for masons, Baliyo Ghar, JICA, and Helvetas will be expanding coverage of OJT. The NRA is also considering potential support from the Indian Government for OJT, through UNDP, in roughly 50 VDCs across Makwanpur and Gorkha. There will still be gaps in the provision of OJT and other POs are requested to focus on providing this training in the districts such as Sindhuli, Makwanpur, and Okhaldhunga where the HRRP has identified huge gaps. There will be a session next Friday, 9th December, in the NRA Conference Hall starting at 2pm on masons training.
- **Retrofitting:** The NRA is moving ahead with the development of a policy for the disbursement of a 100,000 NPRs grant for retrofitting. If there is no TA in place to support this then it is futile or inappropriate to go ahead with the retrofitting grant. At the moment there is no strong offer from POs for this. The government system cannot provide this so it will have to come from POs. There are indications that DFID will make some funds available for POs to deliver TA for retrofitting, and the NRA are very hopeful that this will take place. There is an urgency about this now as households have to take a decision very soon and it will be a lost opportunity if we don't move ahead now. The NRA will not move ahead robustly with retrofitting if the TA to back up the grant is not there. The NRA will be holding a session on retrofitting in the next two weeks and information from the recent HRRP technical session on retrofitting can be found [here](#).

The NRA informed that the previous version of the HRRP had stayed a bit outside of the NRA, but the new version of the HRRP is now working more closely together and an interactive working relationship has been developed. The NRA and the HRRP are now working in harmony with each other and working to transfer this to the districts as well as formalising the relationship.

Tearfund: In Makwanpur, Tearfund are providing the housing grant to 600 households. As the inspection process had not been finalised by the NRA, Tearfund have moved ahead with the inspections and disbursement of the second and third tranches of the grant using partners' engineers to conduct the inspections. Now some households have approached Tearfund to say that they want to return the grant because Tearfund is only providing 250,000 NPRs and they would prefer to receive the grant from the GoN who are providing 300,000 NPRs. Tearfund do not have the resources to cover the additional 50,000 NPRs and are requesting that the NRA cover this. A formal letter requesting this has already been submitted to the NRA.

- One thousand copies of the inspection SOP have been printed, and printing of additional copies is ongoing. Training will be provided to all DL-PIU engineers on the inspection SOP. Even without training, the DL-PIU

engineers should have conducted the inspection visits for the 600 households supported by Tearfund. If POs have any issues with the performance of the DL-PIU engineers they are requested to inform Dr. Shrestha.

- Tearfund are encouraged to find the resources to provide the additional 50,000 NPRs. However, if this is not possible the GoN will provide the top up as it is clear that Tearfund's commitment is for 250,000 NPRs as per their project approval.

NSET: The NSET project approval has been amended to include On the Job Training for masons in all VDCs of Dhading, Dolakha, and Nuwakot. This is to cover any gaps that may exist so where a VDC is already covered for OJT by another PO, NSET will not implement. NSET will be coordinating closely with the HRRP and NRA to ensure that all gaps are filled and duplications avoided.

HRRP: Accurate reporting to the NRA and HRRP is extremely important so that information on coverage of OJT, and other activities, can be collated and analysed to provide strong guidance on where there are gaps or duplications.

Tearfund: Regarding the Build Change mobile app for inspection, what guidelines are followed for data protection?

- This is looked after by the NRA Management Information System division. Once the MIS is fully up and running all inspection data will be integrated through this system.

NSET: The kind of hybrid construction that JICA mentioned in their presentation should be avoided through resolving such issues in the local context. Achieving 100% compliance is going to be very difficult so all organisations need to work together on developing corrective measures to address non-compliances.

UNDP: In retrofitting there are two key issues; non-compliant new construction and partially damaged houses. It is recommended that these issues be considered separately.

UNOPS: Regarding the coverage map for training for existing masons (available [here](#)), as the numbers are based on community perception of the numbers of existing masons in their ward it will be important unpack this and get more detail in order to understand the impact. The estimate for the financial resources required to cover full TA seems a bit low. One area which UNOPS have not seen is follow up after the 7 day training for existing masons and this will be a very costly job. It was also highlighted that trying to complete reconstruction in just 2 years will be difficult, and based on reconstruction experience in other countries it is likely to take longer than that.

BBC Media Action: When will enrolment in the three districts in the Kathmandu valley be completed?

NRA: The Baliyo Ghar target is for a minimum of 70% of households achieving compliance. What support will be provided to make the other 30% of houses safer?

- The programme is aiming to achieve 100% compliance, but the minimum condition is that at least 70% of houses in the working areas will be compliant and disaster resilient. It will be very difficult to influence all house owners so that they fully accept all technical suggestions. However, through technical assistance including supervision and provision of skilled labour, every effort is being made to do this. The 70% mark is a tolerance to measure targets.

Build Change: Unfortunately it is not possible to have 100% of houses compliant. What is required instead is to document how far from compliance houses are, and the level of compliance they reached because of the technical assistance that was provided. At the moment the system has an in built rigidity where compliance is black and white rather than gradient and this is a challenge.

Build Change: We previously approached the NRA regarding an expansion of the existing project approval to expand the organisation's work to Makwanpur. They were told that this was a good idea but that there is no formal process to expand existing project approval. How should they proceed?

- Dr. Shrestha informed that it is not the case that project approval with the NRA cannot be amended, stressing that any agreement with the NRA is amendable. The NRA is currently in the process of amending agreements with Helvetas and NSET to allow for expansion of On the Job training activities by both organisations. POs were

requested to inform Dr. Shrestha (contact details provided below) if they encounter any problems with the amendment of project approvals. In the case of Build Change Dr. Shrestha informed that he will facilitate their expansion into Makwanpur.

Next Steps

- Agree the minimum requirements for TA. POs are requested to provide feedback on this, either generally or with specific reference to the discussion paper available [here](#), to the HRRP Recovery Advisor, Siobhan Kennedy (siobhan@hrrpnepal.org).
- The HRRP is revising the 4W to reflect the minimum requirements for TA and the new 4W format will be used for the next round of data collection which will be submitted from district to national level by the 25th December.
- Partners interested to expand their working areas under the scope of their existing project approval are requested to link directly with the HRRP National Coordinator, Loren Lockwood, on this (loren.lockwood@hrrpnepal.org). The HRRP can facilitate the process with the NRA to amend project approvals for expanded coverage. Partners are requested to focus on expanding coverage of:
 - On the Job Training
 - Technical assistance
- The NRA is considering the potential for a retrofitting grant, but if sufficient technical assistance is not in place this will not be viable. To support the NRA with the planning for this, we would request that any PO with capacity or interest to deliver TA for retrofitting contact the HRRP National Coordinator, Loren Lockwood, on this (loren.lockwood@hrrpnepal.org) on this so that we can develop a comprehensive picture of the potential level of coverage of TA for retrofitting.
- Partners experiencing any difficulties with the DL-PIU engineers can send specific details to the HRRP Recovery Advisor, Siobhan Kennedy (siobhan@hrrpnepal.org), and the HRRP will work with the NRA and MoUD CL-PIU to address the issues.

Reference Documents

- NRA Inspection SOP available [here](#)
- NRA NGO Mobilisation Guidelines available [here](#)
- NRA Community Based Committee directive available [here](#)
- Post Disaster Recovery Framework available [here](#)
- Shelter Cluster Recovery Position Paper available [here](#)
- Information from the Shelter Cluster TWG on technical assistance and training available [here](#)
- HRRP Briefing Pack available [here](#)

Contact Details

Contact details for presenters at the working session are provided below in order to facilitate follow up on any of the topics covered in the session.

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